



OPEN

DATA DESCRIPTOR

A benchmark dataset and baseline methods for rock microstructure interpretation in SEM images

Yao Zhang^{1,2}, Xinming Wu³✉ & Jiachun You⁴

Accurate segmentation of rock microstructures in Scanning Electron Microscope (SEM) images is essential for analyzing porosity and mineral composition. Deep learning (DL) techniques are promising to automate this segmentation, however, their effectiveness is hindered by the lack of labeled SEM datasets. To address this challenge, we developed a standardized SEM dataset including images of mudstone, sandstone, and shale. To improve the quality and diversity of the dataset, we applied preprocessing steps such as magnification standardization, median filtering, and contrast-limited adaptive histogram equalization to each SEM image. **Using this dataset, we compared traditional segmentation methods with state-of-the-art DL models.** Our results demonstrate that DL models significantly outperform traditional methods in capturing complex microstructural details. To facilitate further research, we publicly release the dataset along with implementations of both traditional segmentation algorithms and DL models as benchmarks, providing a valuable reference for further methodological advancements. This study not only offers a high-quality dataset but also contributes to the ongoing development of automated rock SEM image analysis through comprehensive performance evaluations.

Background & Summary

In petroleum exploration, the distribution of pores plays a critical role in determining the oil and gas storage capacity of rocks^{1–3}. Accurately assessing porosity and pore structure not only helps identify optimal drilling locations but also aids in optimizing extraction strategies, thereby maximizing resource recovery and minimizing exploration risks^{4–6}. Additionally, through detailed analysis of pore characteristics, a better understanding of the fluid dynamics within reservoirs can be achieved, enhancing the accuracy of predictive models and consequently improving overall extraction efficiency and economic benefits^{7,8}.

Well logging techniques, such as bulk density, neutron, and sonic logs, can provide estimates of porosity at the log scale. Among these, the bulk density log is often the most reliable for porosity estimation, particularly when mineralogy and fluid properties are well constrained. However, they often lack detailed information on pore morphology, size, and spatial distribution. Additionally, these methods rely on empirical formulas, which may lead to inaccuracies due to variations in formation conditions, especially in multi-mineral mixed reservoirs⁹. Traditional methods, such as core observation and thin section analysis, offer visual insights into pore morphology. However, they are limited by the resolution of optical microscopy, which cannot capture pore details at the micron or nanometer scale^{10,11}. At the microscopic scale, particularly in unconventional reservoirs like tight sandstones and shales, neither logging techniques nor traditional porosity analysis methods can accurately characterize the pore network.

Compared to traditional optical microscopy, Scanning Electron Microscopy (SEM) offers significantly higher resolution, enabling more detailed imaging of pore structures and microtextural features in rock, which is especially important for studying complex pore systems in unconventional reservoirs, such as tight sandstones and shales^{12,13}. Quantitative analysis of SEM images enables precise measurements of parameters such as porosity, pore size distribution, and pore shape¹⁴. Therefore, SEM image analysis of rock samples is an effective tool for

¹State Key Laboratory of Precision Geodesy, School of Earth and Space Sciences, University of Science and Technology of China, Hefei, 230026, China. ²Mengcheng National Geophysical Observatory, University of Science and Technology of China, Mengcheng, 233500, China. ³Institute of Advanced Technology, School of Earth and Space Sciences, University of Science and Technology of China, Hefei, 230026, China. ⁴School of Geophysics, Chengdu University of Technology, Chengdu, 610059, China. ✉e-mail: xinmwu@ustc.edu.cn

	Mudstone	Sandstone	Shale
Sampling Location	Japanese Sea region	North Sea	China
Image Size (pixels)	heights: 813–1770 widths: 1020–2048	heights: 8192 widths: 6144	heights: 680–5866 widths: 1024–6826
Field of View	7.5 μm , 20 μm	1.39 mm	0.83 μm –500 μm
Resolution ($\mu\text{m}/\text{pixel}$)	0.0037–0.0196	0.1697	0.0002–0.6510
Magnification	15,000 $\times$, 40,000 $\times$	—	150 $\times$ –120,000 $\times$

Table 1. Detailed Overview of Collected Rock SEM Image Data.

pore identification¹⁵. Currently, pore identification from SEM images mainly relies on manual interpretation by experts, a process that is both subjective and inefficient, especially for large and complex datasets. The subjectivity and time-consuming nature of the interpretation process reduce the reliability and efficiency of pore analysis in rock SEM images.

The solution lies in developing automated models or tools that ensure objective interpretation while reducing time and costs. Image segmentation methods have been widely used in geoscience to segment geological boundaries or objects from various types of images^{16–18}. Traditional image segmentation algorithms mainly rely on statistical features such as grayscale values and texture for regional segmentation^{19–22}, or on gradient information from grayscale images to identify object boundaries^{23–25}. However, traditional algorithms often require manual parameter settings, such as thresholds, seed points, or initial contours. This can lead to instability in segmentation results, particularly for SEM images characterized by broad pore-size distributions, strong contrast heterogeneity, multi-phase mineral assemblages, and highly anisotropic or irregular pore geometries. In rock SEM images, the boundaries between mineral particles and pores are often unclear and lack strong contrast, making traditional segmentation methods less effective for this task²⁶. Unlike traditional task-specific algorithms, deep learning architectures, which consist of multiple neural network layers, allow computational models to learn data representations. Recent studies have applied deep learning methods to rock SEM image segmentation with promising results^{27–29}. However, the performance of deep learning models depends heavily on the quality and diversity of available datasets^{30,31}. Several public SEM datasets are available, but most lack simultaneous coverage of multiple lithologies with complete pore masks. For example, Becerra *et al.* released a dataset that includes sandstone, shale, and carbonate samples³², however, only the sandstone subset contains pixel-level pore annotations. The “Unconsolidated muds from the Nankai Trough” dataset offers mudstone images with full pore annotations but represents only a single lithology³³. Additionally, the North Sea Sandstone dataset and the Leman Sandstone dataset also focus solely on sandstone and lack pixel-level pore annotations^{34,35}. Therefore, a comprehensive public dataset including multiple lithologies with consistent pixel-level pore annotations remains unavailable. This limitation restricts the widespread application and thorough evaluation of deep learning techniques in rock SEM image segmentation, highlighting the need for a diverse, high-quality rock SEM image dataset.

To address this issue, we create a diverse and representative rock SEM image dataset, consisting of a large number of collected and annotated samples. We also apply a series of preprocessing steps, including magnification standardization, median filtering, and contrast-limited adaptive histogram equalization to improve the diversity and standardization of the dataset. This dataset includes three lithologies: sandstone, shale, and mudstone. By encompassing these key lithologies, the dataset provides a solid foundation for effective training of deep learning models. To evaluate the utility and segmentation performance of the dataset, we applied a series of baseline algorithms. This includes four classical traditional methods: otsu’s method, adaptive mean thresholding, watershed algorithm, and K-means clustering, all of which are known for their simplicity and effectiveness in image segmentation tasks^{19,36–38}. Furthermore, two deep learning models, U-Net and DeepLabv3+, were employed due to their proven effectiveness in semantic segmentation tasks^{39,40}. This study offers a structured framework for the accurate and efficient identification of pores and fractures in rock SEM images, facilitating the precise analysis of oil and gas reservoirs.

Method

Data Collection and Annotation. Our dataset comprises three lithologies-mudstone, sandstone, and shale-classified on compositional and textural criteria that follow standard sedimentological practice^{41–43}. Sandstone is dominated by sand-sized detrital grains in a framework-supported fabric, indicative of relatively high-energy depositional settings. Shale and mudstone are both fine-grained, clay-rich siliciclastics⁴³, but they differ in rock fabric: shale exhibits pronounced fissility and lamination produced by aligned platy clay minerals, which can impart anisotropic mechanical behaviour and, in some cases, hydrocarbon storage capacity^{44,45}. Mudstone, by contrast, is massive, lacks fissility, and is normally non-reservoir in nature. Because visual and terminological overlap between shale and mudstone can cause ambiguity, we keep mudstone as a separate category; this choice also gives future users the flexibility to analyse the two lithologies either independently or in combination, depending on their research needs. This section provides a detailed overview of the data sources, collection process, and other relevant details. Table 1 summarizes the key information of all raw data, while Fig. 1 showcases a selection of the collected raw images.

Mudstone. The mudstone samples used in this study were originally collected from unconsolidated mud in the Kumano Basin of the Nankai Trough, Japan, by the International Ocean Discovery Program (IODP) during Expeditions 315 and 338 at Site C0002, at depths less than 1.1 kilometers below the seafloor³³. The raw SEM

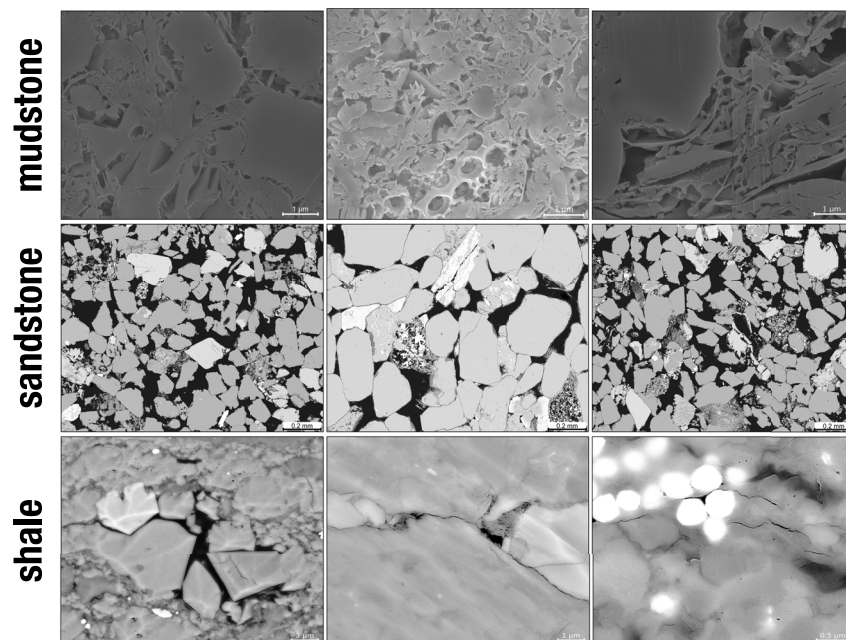


Fig. 1 Representative raw images of three lithologies of mudstone (first row), sandstone (second row), and shale (third row).

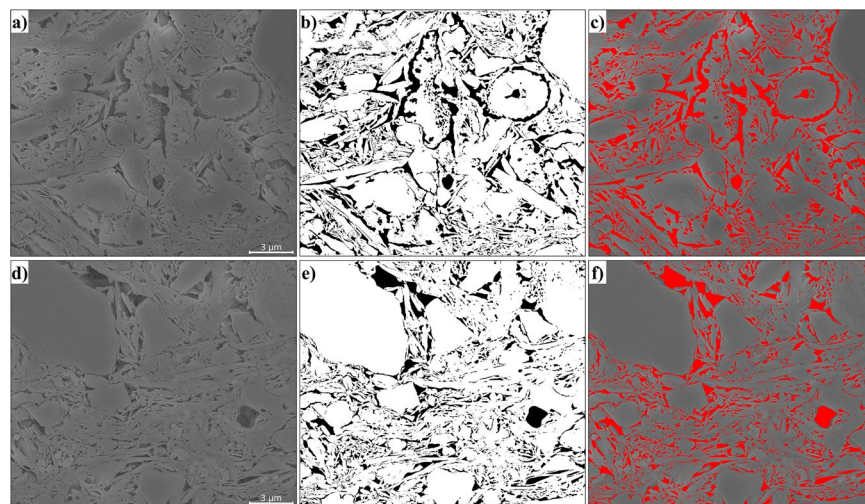


Fig. 2 Original mudstone image and corresponding labels. (a,d) displays the original Scanning Electron Microscope (SEM) image of the mudstone. (b,e) represents the segmentation label corresponding to (a,d), where the black regions indicate the identified pore spaces. To better visualize the location of the pores within the original image, (a) and (b) were overlaid, resulting in image (c), (d) and (e) were overlaid, resulting in image (f). In this overlay, the red regions highlight the pores.

images and their corresponding segmentation labels were obtained from the publicly available Digital Rocks Portal (<https://doi.org/10.17612/P7F59W>)³³. As described by Bihani *et al.*¹⁵, the core surfaces were polished using an Ar-ion cross-section and coated with a 6-nanometer layer of iridium to minimize surface charging effects, and imaged at magnifications of 15,000 $\times$ or 40,000 $\times$, corresponding to horizontal field widths of 20 μm or 7.5 μm , respectively.

In this study, we downloaded this dataset and preprocessed the images. The dataset contained 49 SEM images from five core samples taken at different depths. The details of the preprocessing procedures are provided in the Construction of a Standardized Dataset section. The original images and their corresponding labels are shown in Fig. 2. We calculated that the proportion of black pixels in the labels for this set of mudstone data is 20.38%.

Sandstone. The SEM images of the sandstone samples used in this study were obtained from two publicly available projects: North Sea Sandstone SEM Images (<https://doi.org/10.17612/36F2-8Q45>)³⁴ and Leman

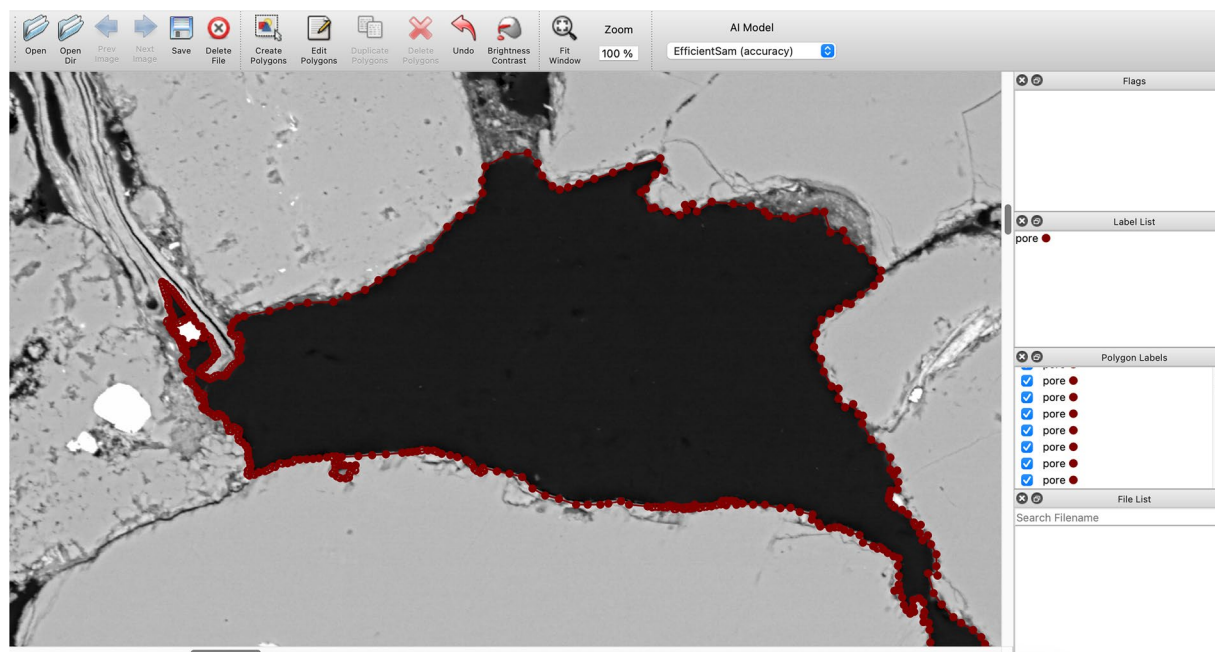


Fig. 3 Manual Annotation of Pores in SEM Image. This figure shows the manual annotation of pores in a rock SEM image using the LabelMe. The red polygons outline the identified pores, capturing their boundaries within the rock matrix. These annotations serve as labels for training machine learning models, supporting automated pore segmentation and analysis.

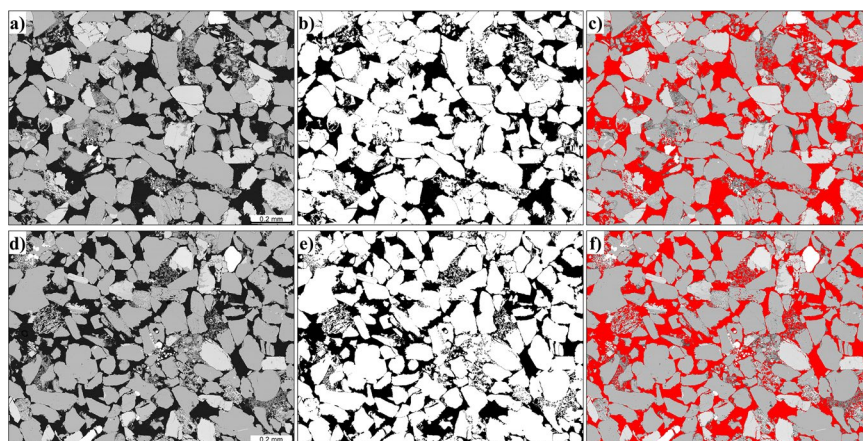


Fig. 4 Sandstone SEM image and corresponding segmentation labels. (a,d) Original SEM image. (b,e) Pixel-wise binary label of pore regions. (c,f) Overlay of the original image and label, red areas indicate the annotated pores.

Sandstone SEM Images (<https://doi.org/10.17612/aky4-9a88>)³⁵. As described in Scott *et al.*⁴⁶, the core samples were collected from the Guillemot A oil field at a depth of approximately 2500 m and the Vanguard oil field at around 2700 m. To prepare the samples, low-density epoxy resin was injected under vacuum, followed by polishing with a diamond abrasive to achieve an optically smooth surface. The polished samples were then coated with a thin layer of carbon graphite under low vacuum to ensure conductivity. The final SEM images have a spatial resolution of $0.17 \mu\text{m}/\text{pixel}$ and a field of view of $1.39 \text{ mm} \times 1.04 \text{ mm}$.

In this work, because the publicly available sandstone images lack pixel-wise annotations of pore structures, we obtained these images and manually annotated the pores using LabelMe⁴⁷, guided by relevant domain knowledge. An example is illustrated in Fig. 3. The original images and their corresponding labels are shown in Fig. 4. We calculated that the proportion of black pixels in the labels for this set of sandstone data is 22.92%.

Shale. The shale dataset used in this study was newly generated and had not been publicly available prior to this work. Samples were collected from six wells in the northeastern part of Shandong Province, China, with burial depths ranging from 3,500 m to 4,200 m. To facilitate clear identification and traceability, each sample was assigned a unique identifier corresponding to its well and depth information. All samples were independently

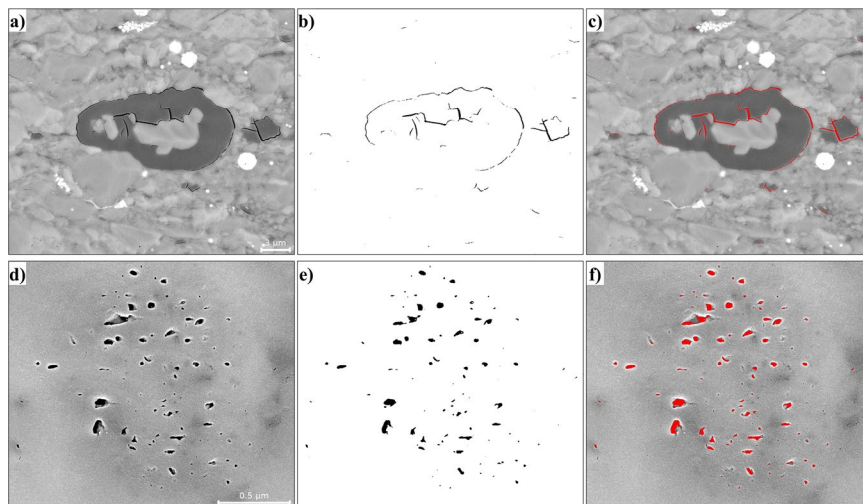


Fig. 5 Shale SEM image and corresponding segmentation labels. (a,d) Original SEM image. (b,e) Pixel-wise binary label of pore regions. (c,f) Overlay of the original image and label, red areas indicate the annotated pores.

prepared, imaged, and manually annotated, covering a variety of lithofacies, including organic-rich laminated calcareous shale with sparry calcite cement and organic-rich laminated calcareous shale rich in micritic calcite. During sample preparation, to minimize polishing scratches and surface contamination, all specimens were subjected to argon-ion polishing under a standardized laboratory protocol, ensuring surface quality consistency and comparability.

SEM imaging was performed on multiple laboratory platforms, including instruments equipped with the ZEISS Crossbeam 550 series. The dataset encompasses multiple imaging modes, such as secondary electron (SE) and backscattered electron (BSE) imaging, and employs detectors including SE2, ETD, BSED, and LFD. Imaging parameters span a wide range, with magnifications from $150\times$ to $120,000\times$, accelerating voltages from 1 kV to 25 kV, and horizontal field widths from $500\text{ }\mu\text{m}$ (low magnification) to $0.83\text{ }\mu\text{m}$ (high magnification). Each SEM image filename includes the well name, file name, magnification, scale bar, and depth, ensuring standardized data management.

In total, 1,119 images were obtained from different wells and depths, encompassing 109 distinct magnification levels. The original images and their corresponding segmentation labels are shown in Fig. 5. The proportion of black pixels in the labels for this shale dataset is 2.50%.

Across the three lithologies, certain SEM images display quality issues, including curtaining artifacts, charging effects, low contrast, and defocus. These imperfections primarily stem from limitations in sample preparation and imaging conditions associated with the original data sources. To reduce their impact and enhance the consistency and overall quality of the dataset, a standardized preprocessing pipeline was applied to all images prior to segmentation and analysis. This workflow was designed to improve image contrast, suppress noise, and retain essential microstructural features. A comprehensive description of the preprocessing procedure is provided in the “Construction of a Standardized Dataset” section.

Construction of a Standardized Dataset. To ensure dataset consistency and facilitate fair comparison in subsequent analyses, all SEM images were resampled at the pixel level to match their nominal magnifications (this operation alters only the image matrix and does not affect the physical field of view or pixel size). Mudstone images recorded at $15,000\times$ are upsampled to $40,000\times$ using nearest-neighbour interpolation, whereas those already at $40,000\times$ are left unchanged. Sandstone images were acquired at a single magnification and therefore required no resampling. Shale images, by contrast, spanned 109 different magnifications. After examining their frequency distribution, we selected the seven levels with the largest image counts, namely $1,000\times$, $2,000\times$, $5,000\times$, $10,000\times$, $30,000\times$, $100,000\times$ and $120,000\times$, and resampled each image to the nearest level ($<1,000 \rightarrow 1,000$; $1,000\text{--}2,000 \rightarrow 2,000$; $2,000\text{--}5,000 \rightarrow 5,000$; $5,000\text{--}20,000 \rightarrow 10,000$; $20,000\text{--}50,000 \rightarrow 30,000$; $50,000\text{--}120,000 \rightarrow 100,000$). This strategy equalises the visual scale across the dataset and preserves the underlying spatial resolution, thereby improving the consistency and comparability of all subsequent image-processing steps.

After resampling all SEM images to a uniform apparent scale, we applied median filtering to remove outliers⁴⁸. This technique replaces each pixel value with the median of its neighboring pixels, effectively eliminating local anomalies such as salt-and-pepper noise. By reducing such artefacts, we minimize the risk of introducing errors during subsequent normalization and neural network training. This step further enhances the quality of the image data, ensuring that the network can more effectively extract and learn relevant features.

Owing to inconsistencies in the imaging process, the SEM images displayed pronounced brightness and contrast variations⁴⁹. To enhance local contrast without amplifying noise, we applied Contrast-Limited Adaptive Histogram Equalization (CLAHE) as a preprocessing step⁵⁰. Fig. 6 compares grayscale histograms of three image sets before and after CLAHE. The post-processing histograms are broader and flatter, confirming a more balanced pixel-value distribution and improved contrast. Consequently, overall image visibility is enhanced and fine details become clearer.

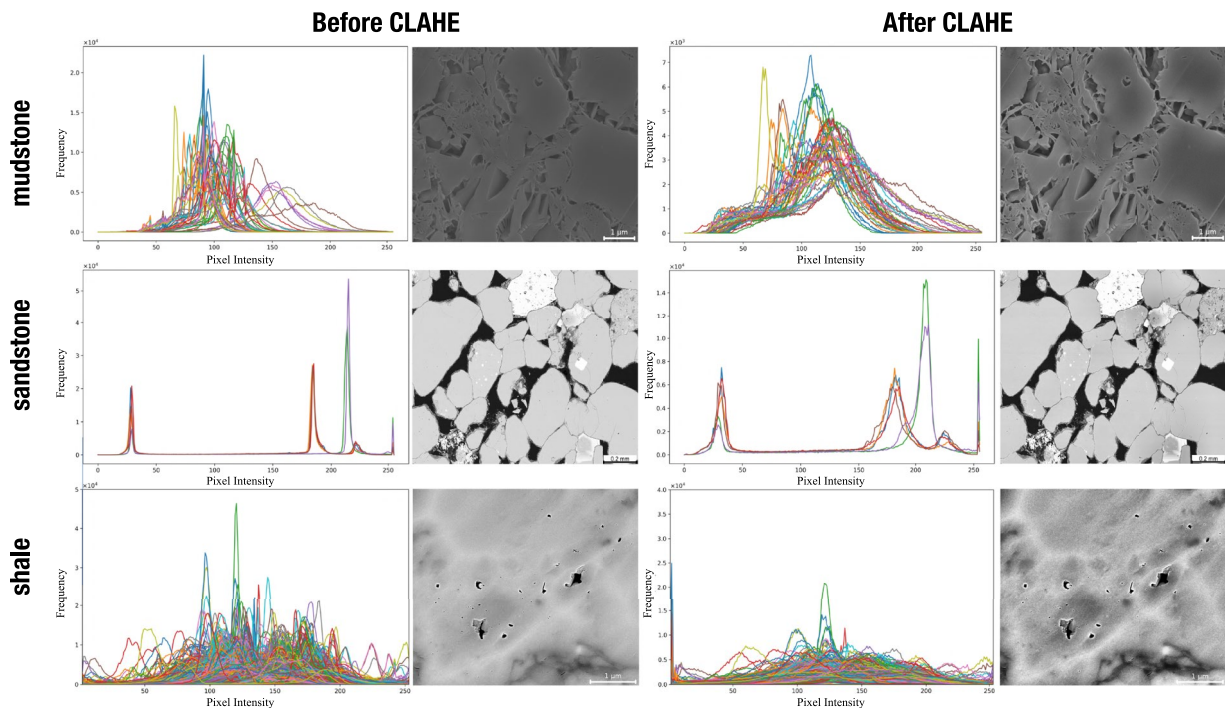


Fig. 6 Comparison of grayscale histograms and image results before and after applying Contrast-Limited Adaptive Histogram Equalization (CLAHE). The left panels represent the grayscale histogram and an example image before CLAHE, while the right panels show the corresponding results after CLAHE enhancement.

	Mudstone	Sandstone	Shale
Label Description	Black pixels indicate pores, white pixels indicate background	Black pixels indicate pores, white pixels indicate background	Black pixels indicate pores, white pixels indicate background
Image Size (pixels)	512 × 512	512 × 512	512 × 512
Field of View (μm)	1.89~3.79	87.04	0.41~500.38
Resolution (μm/pixel)	0.0037~0.0074	0.1700	0.0008~0.9773
Magnification	40,000×	—	1,000×, 2,000×, 5,000×, 10,000×, 30,000×, 100,000×, 120,000×
Count	2477	1152	11325

Table 2. Detailed Overview of the Final Processed Rock SEM Dataset.

Subsequently, to maintain consistent image size and increase the total number of images, all images were cropped to a size of 512 × 512 pixels. Images with a corresponding porosity of less than 0.1%, which were essentially blank, were removed. The final dataset consists of **2,477** mudstone images, **1,152** sandstone images, and **11,325** shale images. With this step, the dataset construction was completed, and we refer to the resulting dataset as **cigRockSEM**. Table 2 provides detailed information about the finalized constructed dataset.

Data Records

The **cigRockSEM** dataset is available at Zenodo (<https://doi.org/10.5281/zenodo.14988631>)⁵¹. After downloading and extracting the archive, a root directory named **data** is created, containing three subdirectories corresponding to the three lithologies: **mudstone**, **sandstone**, and **shale**. Each subdirectory contains two folders, **imgs** and **masks**, where **imgs** stores the pre-processed SEM images in PNG format with a resolution of 512 × 512 pixels, and **masks** stores the corresponding segmentation-label files in PNG format. The filenames in **imgs** and **masks** are identical, enabling direct one-to-one pairing between each image and its ground-truth mask.

Technical Validation

After constructing the dataset, we evaluated four traditional image segmentation algorithms and two deep learning-based semantic segmentation methods to thoroughly assess their performance in the subsequent experimental phase.

Benchmarking Traditional Image Segmentation Methods on cigRockSEM. *Otsu’s method.* Otsu’s method is a global thresholding technique that selects an optimal threshold by minimizing the intra-class variance of grayscale pixel intensities. It assumes that the image contains two distinct pixel classes,

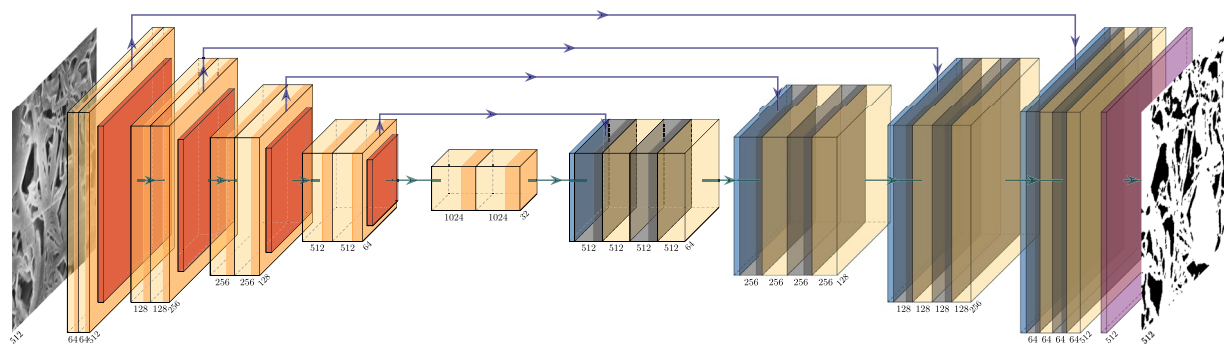


Fig. 7 Schematic Diagram of the U-Net Architecture³⁹.

Name	Parameter
GPU Model	Tesla V100-PCIE-32GB
Memory	32GB
Operating System	Linux
Programming Language	Python 3.9
Deep Learning Framework	Pytorch
Loss Function	Cross Entropy Loss
Optimizer	RMSprop
Epoch	100
Batch Size	16

Table 3. Hardware and Software Configuration for Training U-Net Model.

typically foreground and background, and performs best when the grayscale histogram shows a clear bimodal distribution¹⁹.

In this study, Otsu’s method was applied directly to the grayscale SEM images in the cigRockSEM dataset without any additional preprocessing. It served as a simple baseline for evaluating segmentation performance. For sandstone samples, where pores typically exhibit strong contrast against a relatively uniform matrix, this method yielded reasonable results. However, its effectiveness decreased significantly in shale and mudstone images, where grayscale histograms are often more complex and lack a distinct separation between pore and non-pore regions.

Adaptive Mean Thresholding. Adaptive mean thresholding is a local binarization method that determines the threshold for each pixel based on the average grayscale intensity within a surrounding neighborhood. It is designed to adapt to local variations in intensity and is often used in images with uneven lighting or complex texture³⁶.

In this study, we applied adaptive mean thresholding to the grayscale SEM images in the cigRockSEM dataset as a comparative method alongside global thresholding. Although it introduces local adaptability, the segmentation results were generally comparable to those obtained using Otsu’s method. In many cases, the method exhibited sensitivity to noise and produced fragmented regions in areas with weak contrast or structural complexity.

Watershed Algorithm. The watershed algorithm is a classical image segmentation technique derived from topographic analysis. In this approach, a grayscale image is interpreted as a three-dimensional relief surface, where higher pixel intensities correspond to elevated regions and lower intensities represent valleys. The algorithm identifies local minima as initial catchment basins and simulates a flooding process. As the simulated water level rises, adjacent basins expand until they intersect, forming watershed lines that delineate region boundaries³⁷.

In this study, the watershed algorithm was applied to the cigRockSEM dataset to segment pore structures from grayscale SEM images. The method showed reasonable effectiveness in segmenting well-defined pore spaces, particularly in sandstone samples with distinct morphological boundaries. However, in fine-grained lithologies such as mudstone and shale, where textures are more heterogeneous and pore boundaries less discernible, the algorithm often resulted in significant over-segmentation. This is primarily due to its sensitivity to local grayscale variations and image noise, which limit its applicability in such contexts.

K-means Clustering. K-means clustering is an unsupervised learning algorithm that partitions data into K clusters by minimizing the within-cluster sum of squared distances to the cluster centroids. In image segmentation tasks, each pixel is treated as a data point in a low-dimensional feature space, typically based on grayscale

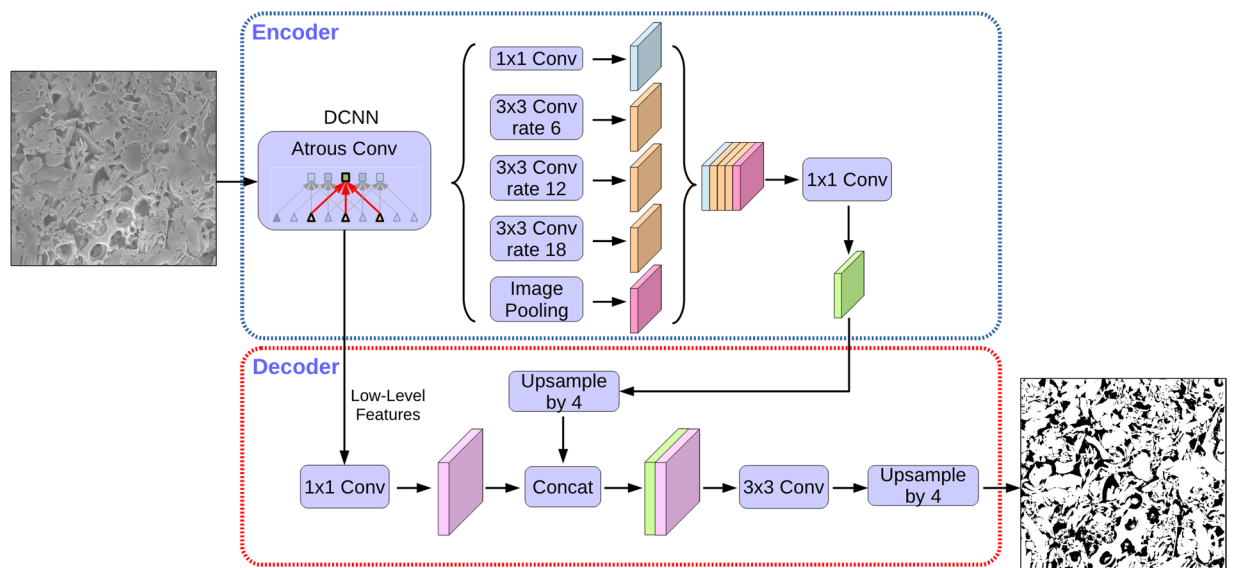


Fig. 8 Schematic Diagram of the DeepLabv3+ Architecture⁴⁰.

Name	Parameter
GPU Model	Tesla V100-PCIE-32GB
Memory	32GB
Operating System	Linux
Programming Language	Python 3.9
Deep Learning Framework	PyTorch
Backbone	ResNet-101
Loss Function	Cross Entropy Loss
Optimizer	Stochastic Gradient Descent
Epoch	100
Batch Size	16

Table 4. Hardware and Software Configuration for Training DeepLabv3+ Model.

or color intensity. In this study, we applied K-means clustering with $K = 2$ to the grayscale SEM images in the cigRockSEM dataset to separate pore and non-pore regions. Each image was reshaped into a one-dimensional array of pixel intensities, and clustering was performed using the standard implementation provided by scikit-learn. After clustering, the cluster with the lower average intensity was identified as the pore phase³⁸.

K-means produced relatively clean segmentation results in high-contrast sandstone images, where pore regions exhibit intensities clearly distinct from the surrounding matrix. However, in shale and mudstone samples, overlapping grayscale distributions and weak contrast often led to inaccurate classification and poorly defined boundaries. The method also showed sensitivity to initialization and lacked spatial context, which further limited its performance in complex rock microstructures.

Benchmarking Deep Learning Methods on cigRockSEM. *U-Net.* U-Net is a widely used convolutional neural network (CNN) architecture³⁹, featuring a symmetric encoder-decoder structure. This architecture consists of two parts: the encoder extracts deep features from the input image, while the decoder combines multi-scale features from the encoder to produce the final segmentation output. This design effectively integrates features at different scales, enabling the model to perform well in image segmentation tasks. The specific architecture used in this study is shown in Fig. 7.

The U-Net model was trained using the dataset constructed in this study. The training and testing sets were split in a ratio of 8:2. The specific parameters used during training, as well as the platform used, are shown in Table 3. After training, the segmentation performance of the model was evaluated on the test set.

DeepLabv3+. The DeepLabv3+ model, proposed by the Google Research team, is an advanced semantic segmentation model⁴⁰. DeepLabv3+ shares a similar Encoder-Decoder architecture with the U-Net model. Its encoder typically uses a pre-trained convolutional neural network (such as ResNet or Xception) as a feature extractor, in this study, we used ResNet-101 as the backbone, which effectively captures high-level semantic information from the input image. Building on this, DeepLabv3+ incorporates the Atrous Spatial Pyramid Pooling (ASPP) module, which extracts multi-scale features using convolutional filters with different dilation

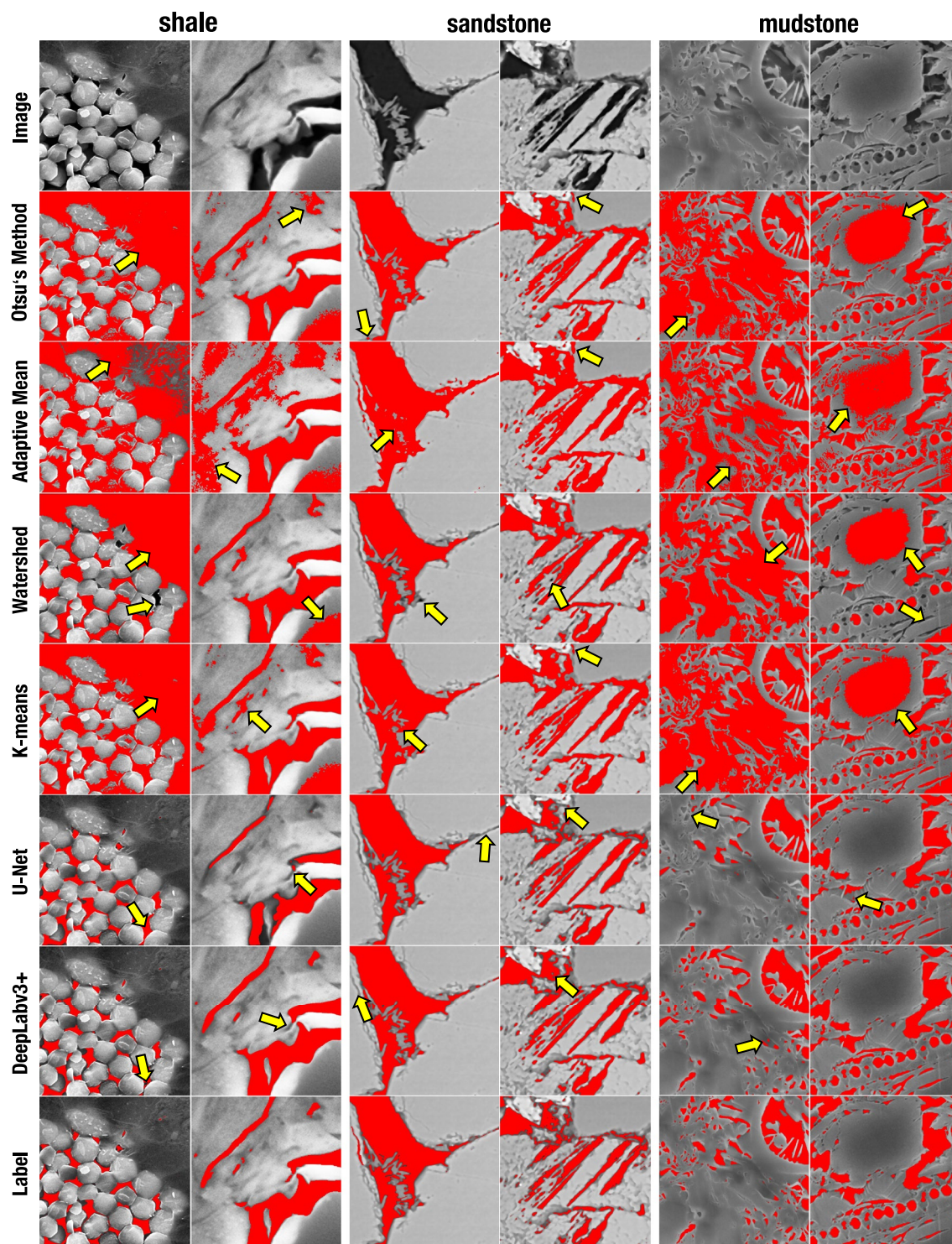


Fig. 9 Comparison of segmentation results using different methods on SEM images of rock samples. The first row shows the original images, with shale in the first two columns, sandstone in the next two, and mudstone in the last two. The second to seventh rows display the segmentation results using Otsu's method, adaptive mean thresholding, watershed algorithm, K-means clustering, U-Net, and DeepLabv3+, respectively. The red regions highlight the detected pore structures. The eighth row shows the corresponding ground truth labels.

rates. This enhances the model's ability to capture context at various scales. Additionally, the model's decoder progressively restores spatial resolution, improving segmentation boundary accuracy. The structure of DeepLabv3+ used in this study is shown in Fig. 8.

The DeepLabv3+ model was trained using the dataset constructed in this study. The training and testing sets were split in a ratio of 8:2. The specific parameters used during the training process, as well as the platform

		Otsu's method	Adaptive Mean Thresholding	Watershed Algorithm	K-means Clustering	U-Net	DeepLabv3+
Mudstone	acc	0.8011	0.6846	0.8189	0.8044	0.9267	0.9317
	F1	0.8514	0.7589	0.8706	0.8536	0.9522	0.9223
	Recall	0.7951	0.6359	0.8364	0.7998	0.9715	0.9318
	mIoU	0.7637	0.6159	0.7879	0.7677	0.7940	0.8583
Sandstone	acc	0.9247	0.8923	0.9346	0.9149	0.9880	0.9837
	F1	0.9443	0.9243	0.9524	0.9325	0.9910	0.9781
	Recall	0.9068	0.8671	0.9325	0.8972	0.9905	0.9808
	mIoU	0.9045	0.8647	0.9174	0.8948	0.9272	0.9573
Shale	acc	0.5758	0.5624	0.6400	0.5729	0.9753	0.9809
	F1	0.7137	0.7089	0.7656	0.7103	0.9769	0.8488
	Recall	0.5684	0.5532	0.6366	0.5652	0.9936	0.8507
	mIoU	0.5656	0.5506	0.6320	0.5625	0.6358	0.7639

Table 5. Segmentation Performance Comparison for Traditional and Deep Learning Methods.

employed, are shown in Table 4. After training, the segmentation performance of the model was evaluated on the test set.

To provide a comprehensive comparison of the performance of four traditional image segmentation algorithms (Otsu's method, adaptive mean thresholding, watershed algorithm, and K-means clustering) and two deep learning-based segmentation methods (U-Net and DeepLabv3+) on various rock SEM images, systematic tests are conducted on an identical set of rock SEM images. The segmentation results of these six methods for mudstone, sandstone, and shale samples are presented in Fig. 9. The red regions highlight the identified pore areas, providing a clear visual basis for comparing the performance of these methods. As illustrated, traditional algorithms, including Otsu's thresholding, adaptive mean thresholding, the watershed algorithm, and K-means clustering, yield comparatively better performance on sandstone, where pore geometry is relatively simple, but still fail to delineate small pores and precise boundaries. Their effectiveness drops further for mudstone and shale, where pore networks are more intricate. Although other image segmentation techniques exist, most require manual tuning of multiple parameters or involve complex iterative processes, limiting their practicality for fast and accurate segmentation of rock SEM images. In contrast, the deep-learning models deliver superior results across all three lithologies, clearly demonstrating a greater capacity to handle complex pore structures.

To enable a fair comparison, we quantitatively evaluate four traditional algorithms (Otsu, adaptive mean thresholding, watershed, and K-means) and two deep-learning models (U-Net and DeepLabv3+). The evaluation metrics include accuracy, recall, F1 score, and mean intersection over union (mIoU), as summarised in Table 5. Accuracy denotes the proportion of correctly classified pixels; recall measures how completely each class (pore or background) is retrieved; the F1-score is the harmonic mean of precision and recall; and mIoU is calculated by averaging the Intersection over Union (IoU) for each class. IoU is the ratio of the intersection of predicted and true regions to the union of these regions. In our dataset, due to the class imbalance, the pore class typically occupies a small portion of the image, which results in a lower IoU for the pore class and a higher IoU for the background class. By using mIoU, we obtain a single value that provides a comprehensive reflection of the model's performance across both categories, offering a more balanced representation of segmentation consistency. Across the three lithologies the convolutional models consistently outperform the classical algorithms. In sandstone, where pore geometry is comparatively simple, classical methods already achieve accuracies above 0.93, yet DeepLabv3+ further increases mIoU to 0.96, indicating sharper boundary delineation. In mudstone, DeepLabv3+ reaches an mIoU of 0.86, approximately eight percentage points higher than the best classical algorithm, while U-Net produces the highest F1-score (0.95). For shale, the most challenging lithology, the best classical mIoU falls below 0.64; DeepLabv3+ maintains 0.76 and U-Net retains an F1-score close to 0.97, highlighting the robustness of convolutional networks to strongly anisotropic and irregular pore systems. Overall, the deep-learning models not only deliver higher average scores but also exhibited greater robustness to lithological variability, markedly improving fine-scale pore detection and the global consistency of the segmentation in samples with complex microstructures.

All evaluations and comparisons in this study are based on manually annotated ground truth labels. While the annotation process may be influenced by human subjectivity, we make efforts to follow standardized labeling protocols to minimize potential bias. The cigRockSEM dataset holds significant potential for both academic research and industrial applications. It can be used to train deep learning models for automated pore detection, support statistical analysis of microstructural pore characteristics, and serve as a pretraining foundation for transfer learning across various geological settings. Nonetheless, there remains substantial room for improvement. In future work, we plan to enhance the cigRockSEM dataset along several dimensions. First, the inclusion of additional reservoir lithologies, such as igneous or metamorphic rocks, could improve the dataset's geological diversity and representativeness. Second, leveraging more advanced SEM systems and adopting standardized imaging parameters could further improve image quality and consistency. Third, increasing the number of high-quality annotated images would contribute to better model training and generalization performance. It is also worth noting that the current dataset only includes binary pore annotations. Future versions could expand the semantic scope to include mineral phases, lithological textures, or structural components, thereby supporting a broader range of geoscientific and machine learning tasks. We believe these enhancements will

significantly improve the scientific utility and application value of the cigRockSEM dataset in both geoscience and AI-assisted rock analysis.

Data availability

The cigRockSEM dataset used in this study is publicly available at Zenodo: <https://doi.org/10.5281/zenodo.14988631>. The dataset is provided in PNG format. Additional details on the dataset contents, file structure, and variable descriptions are provided in the Data Records section.

Code availability

The code for deep learning training and inference, as well as the implementation of traditional methods for benchmarking, is available at https://github.com/Zhangyao2414/cigRockSEM_dataset.

Received: 11 March 2025; Accepted: 4 September 2025;

Published: 22 October 2025

References

- Zou, C. *et al.* Global importance of ‘continuous’ petroleum reservoirs: accumulation, distribution and evaluation. *Pet. Explor. Dev.* **36**, 669–682, [https://doi.org/10.1016/S1876-3804\(10\)60001-8](https://doi.org/10.1016/S1876-3804(10)60001-8) (2009).
- Hao, F., Zou, H. & Lu, Y. Mechanisms of shale gas storage: implications for shale gas exploration in China. *AAPG Bull.* **97**, 1325–1346, <https://doi.org/10.1306/02141312091> (2013).
- Zhu, H. *et al.* Impact of tectonism on pore type and pore structure evolution in organic-rich shale: implications for gas storage and migration pathways in naturally deformed rocks. *Fuel* **228**, 272–289, <https://doi.org/10.1016/j.fuel.2018.04.137> (2018).
- Prensky, S. E. Advances in borehole imaging technology and applications. *Geol. Soc. London Spec. Publ.* **159**, 1–43, <https://doi.org/10.1144/GSL.SP.1999.159.01.01> (1999).
- Wang, L. *et al.* Advances in improved/enhanced oil recovery technologies for tight and shale reservoirs. *Fuel* **210**, 425–445, <https://doi.org/10.1016/j.fuel.2017.08.095> (2017).
- Qin, L. *et al.* Evolution of the pore structure in coal subjected to freeze-thaw using liquid nitrogen to enhance coalbed methane extraction. *J. Pet. Sci. Eng.* **175**, 129–139, <https://doi.org/10.1016/j.petrol.2018.12.037> (2019).
- Bohacs, K. M. *et al.* The spectrum of fine-grained reservoirs from ‘shale gas’ to ‘shale oil’/tight liquids: essential attributes, key controls, practical characterization. In *Proc. Int. Petrol. Technol. Conf. IPTC-16676* 1–16 (2013).
- Wang, Y. D. *et al.* Deep learning in pore scale imaging and modeling. *Earth-Sci. Rev.* **215**, 103555, <https://doi.org/10.1016/j.earscirev.2021.103555> (2021).
- Rider, M. H. *The Geological Interpretation of Well Logs*. Blackie Academic & Professional, Glasgow (1986).
- Schmitt, D. R. Geophysical properties of the near surface Earth: seismic properties. In *Treatise on Geophysics* (2nd edn) (ed. Schubert, G.) vol. 11, 43–87, <https://doi.org/10.1016/B978-0-444-53802-4.00190-1> (Elsevier, 2015).
- Curtis, M. E., Ambrose, R. J. & Sondergeld, C. H. *et al.* Structural characterization of gas shales on the micro- and nano-scales. In *Proc. Can. Unconv. Resour. Int. Petrol. Conf. SPE-137693-MS* <https://doi.org/10.2118/137693-MS> (Calgary, 2010).
- Curtis, M. E. *et al.* Microstructural investigation of gas shales in two and three dimensions using nanometer-scale resolution imaging. *AAPG Bull.* **96**, 665–677, <https://doi.org/10.1306/08151110188> (2012).
- Loucks, R. G., Reed, R. M. & Ruppel, S. C. *et al.* Morphology, genesis and distribution of nanometer-scale pores in siliceous mudstones of the Mississippian Barnett Shale. *J. Sediment. Res.* **79**, 848–861, <https://doi.org/10.2110/jsr.2009.092> (2009).
- Klaver, J., Desbois, G. & Urai, J. L. *et al.* BIB-SEM study of the pore space morphology in early mature Posidonia Shale from the Hils area, Germany. *Int. J. Coal Geol.* **103**, 12–25, <https://doi.org/10.1016/j.coal.2012.06.012> (2012).
- Bihani, A., Daigle, H. & Santos, J. E. *et al.* MudrockNet: semantic segmentation of mudrock SEM images through deep learning. *Comput. Geosci.* **158**, 104952, <https://doi.org/10.1016/j.cageo.2021.104952> (2022).
- Andrew, M. A. quantified study of segmentation techniques on synthetic geological XRM and FIB-SEM images. *Comput. Geosci.* **22**, 1503–1512, <https://doi.org/10.1007/s10596-018-9768-y> (2018).
- Radoux, J. *et al.* Improving ecotope segmentation by combining topographic and spectral data. *Remote Sens.* **11**, 354 (2019).
- Fang, C. *et al.* A globally distributed dataset of coseismic landslide mapping via multi-source high-resolution remote sensing images. *Earth Syst. Sci. Data* **16**, 4817–4842, <https://doi.org/10.5194/essd-16-4817-2024> (2024).
- Otsu, N. A threshold selection method from gray-level histograms. *IEEE Trans. Syst. Man Cybern.* **9**, 62–66, <https://doi.org/10.1109/TSMC.1979.4310076> (1979).
- Pun, T. A new method for grey-level picture thresholding using the entropy of the histogram. *Signal Process.* **2**, 223–237 (1980).
- Tremeau, A. & Borel, N. A region growing and merging algorithm to color segmentation. *Pattern Recognit.* **30**, 1191–1203 (1997).
- Mingireanov Filho, I. *et al.* Segmentation of sandstone thin section images with separation of touching grains using optimum path forest operators. *Comput. Geosci.* **57**, 146–157, <https://doi.org/10.1016/j.cageo.2013.04.011> (2013).
- Kanopoulos, N., Vasanthavada, N. & Baker, R. L. Design of an image edge detection filter using the Sobel operator. *IEEE J. Solid-State Circuits* **23**, 358–367, <https://doi.org/10.1109/4.996> (1988).
- Rosenfeld, A. The max Roberts operator is a Hueckel-type edge detector. *IEEE Trans. Pattern Anal. Mach. Intell.* **3**, 101–103 (1981).
- Yang, L., Wu, X. & Zhao, D. *et al.* An improved Prewitt algorithm for edge detection based on noised image. In *Proc. 4th Int. Congr. Image Signal Process. (CISP)* vol. 3, 1197–1200 <https://doi.org/10.1109/CISP.2011.6100495> (Shanghai, 2011).
- Chen, Z. *et al.* Deep learning-based method for SEM image segmentation in mineral characterization: an example from Duvernay Shale samples in the Western Canada Sedimentary Basin. *Comput. Geosci.* **138**, 104450, <https://doi.org/10.1016/j.cageo.2020.104450> (2020).
- Karimpouli, S. & Tahmasebi, P. Segmentation of digital rock images using deep convolutional autoencoder networks. *Comput. Geosci.* **126**, 142–150, <https://doi.org/10.1016/j.cageo.2019.02.003> (2019).
- Niu, Y. *et al.* Digital rock segmentation for petrophysical analysis with reduced user bias using convolutional neural networks. *Water Resour. Res.* **56**, e2019WR026597, <https://doi.org/10.1029/2019WR026597> (2020).
- Balciewicz, M. *et al.* Digital rock physics: a geological driven workflow for the segmentation of anisotropic Ruhr sandstone. *Front. Earth Sci.* **9**, 673753, <https://doi.org/10.3389/feart.2021.673753> (2021).
- Varfolomeev, I., Yakimchuk, I. & Safonov, I. An application of deep neural networks for segmentation of microtomographic images of rock samples. *Computers* **8**, 72, <https://doi.org/10.3390/computers8040072> (2019).
- Ma, Z. *et al.* Enhancing rock image segmentation in digital rock physics: a fusion of generative AI and state-of-the-art neural networks. *Preprint at* <https://arxiv.org/abs/2311.06079> (2023).
- Becerra, D. *et al.* Generating a labeled data set to train machine learning algorithms for lithologic classification of drill cuttings. *Interpretation* **10**, SE85–SE100, <https://doi.org/10.1190/INT-2021-0194.1> (2022).
- Milliken, K. L., Prodanovic, M., Nole, M. & Daigle, H. Unconsolidated muds from the Nankai Trough (dataset). *Digital Rocks Portal* <https://doi.org/10.17612/P7F59W> (2025).

34. Scott, G. North Sea Sandstone SEM Images (dataset). *Digital Rocks Portal* <https://doi.org/10.17612/36F2-8Q45> (2025).
35. Scott, G. Leman Sandstone SEM Images (dataset). *Digital Rocks Portal* <https://doi.org/10.17612/aky4-9a88> (2025).
36. Jähne, B. *Digital Image Processing* 6th edn (Springer, Berlin, 2005).
37. Vincent, L. & Soille, P. Watersheds in digital spaces: an efficient algorithm based on immersion simulations. *IEEE Trans. Pattern Anal. Mach. Intell.* **13**, 583–598, <https://doi.org/10.1109/34.87344> (1991).
38. MacQueen, J. Some methods for classification and analysis of multivariate observations. In *Proc. 5th Berkeley Symp. Math. Stat. Probab.* vol. 1, 281–298 (Univ. California Press, Berkeley, 1967).
39. Ronneberger, O., Fischer, P. & Brox, T. U-net: convolutional networks for biomedical image segmentation. In *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)* vol. 9351, 234–241, https://doi.org/10.1007/978-3-319-24574-4_28 (Springer, Cham, 2015).
40. Chen, L.-C., Zhu, Y., Papandreou, G., Schroff, F. & Adam, H. Encoder-decoder with atrous separable convolution for semantic image segmentation. In *Proc. Eur. Conf. Comput. Vis. (ECCV)* vol. 11211, 801–818, https://doi.org/10.1007/978-3-030-01234-2_49 (Springer, Cham, 2018).
41. Aplin, A. C. & Macquaker, J. H. S. Mudstone diversity: origin and implications for source, seal and reservoir properties in petroleum systems. *AAPG Bull.* **95**, 2031–2059, <https://doi.org/10.1306/03281110162> (2011).
42. Berg, R. R. Reservoir morphologies for Gulf Coast turbidite sandstones, Texas. *AAPG Bull.* **63**, 418–419 (1979).
43. Boggs, S. Jr. *Principles of Sedimentology and Stratigraphy* 5th edn (Pearson Prentice Hall, Upper Saddle River, NJ, 2012).
44. Curtis, J. B. Fractured shale-gas systems. *AAPG Bull.* **86**, 1921–1938, <https://doi.org/10.1306/61EEDDBE-173E-11D7-8645000102C1865D> (2002).
45. Passey, Q. R., Bohacs, K. M. & Esch, W. L. *et al.* From oil-prone source rock to gas-producing shale reservoir - geologic and petrophysical characterization of unconventional shale-gas reservoirs. In *Proc. SPE Int. Oil Gas Conf. Exhib. China SPE-131350-MS* <https://doi.org/10.2118/131350-MS> (Beijing, 2010).
46. Scott, G., Wu, K. & Zhou, Y. Multi-scale image-based pore space characterisation and pore network generation: case study of a North Sea sandstone reservoir. *Transp. Porous Media* **129**, 855–884 (2019).
47. Russell, B. C., Torralba, A., Murphy, K. P. & Freeman, W. T. LabelMe: a database and web-based tool for image annotation. *Int. J. Comput. Vis.* **77**, 157–173, <https://doi.org/10.1007/s11263-007-0090-8> (2008).
48. Jain, A. K. *Fundamentals of Digital Image Processing*. (Prentice-Hall, Englewood Cliffs, NJ, 1989).
49. Ali, A., Zhang, N. & Santos, R. M. Mineral characterization using scanning electron microscopy (SEM): a review of the fundamentals, advancements and research directions. *Appl. Sci.* **13**, 12600, <https://doi.org/10.3390/app132312600> (2023).
50. Reza, A. M. Realization of the contrast limited adaptive histogram equalization (CLAHE) for real-time image enhancement. *J. VLSI Signal Process. Syst.* **38**, 35–44, <https://doi.org/10.1023/B:VLSI.0000028532.53893.82> (2004).
51. Zhang, Y. & Wu, X. cigRockSEM: a benchmark dataset and baseline methods for rock microstructure interpretation in scanning electron microscope (SEM) images (Version 1.0.0). *Zenodo* <https://doi.org/10.5281/zenodo.14988631> (2025).

Acknowledgements

This research is financially supported by the National Science Foundation of China under grant 42374127. The authors thank the USTC supercomputing center for providing computational resources for this project. We would also like to express our sincere gratitude to Dr. Elizabeth Miller, the handling editor, for her professional guidance and constructive comments. Furthermore, we greatly appreciate the insightful suggestions and critical feedback provided by the anonymous reviewers, which have significantly improved the quality of this work.

Author contributions

Yao Zhang wrote the manuscript and developed the Python code for data preprocessing and traditional image segmentation algorithms. Xinming Wu was involved in conceptualisation and manuscript preparation. Jiachun You supported the work and provided suggestions for the draft improvement.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to X.W.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025, modified publication 2026